

SION ALIN-MIHĂIȚĂ

AI Software Engineer

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PROFESSIONAL SUMMARY

I am a student at Alexandru Ioan Cuza University of Iași, passionate about programming and building things through it. I enjoy learning and working with Python, machine learning, computer vision, large language models, and AI-powered applications. Deeply committed to continuous growth and open to new opportunities that push my problem-solving abilities and help me grow.

CORE COMPETENCIES

- **AI & Machine Learning:** Computer Vision, Deep Learning, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Convolutional Neural Networks (CNNs), Object Detection, Support Vector Machines (SVM), NLP & AI Agents.
- **Software Engineering:** Object-Oriented Programming (OOP), Data Structures & Algorithms, Mathematical Modeling, System Design, One-vs-All Classification Strategies.
- **Tools & Methodologies:** Linux Environment (Mint), Version Control (Git), Agile Workflows, Debugging & Problem Solving.

KEY TECHNICAL PROJECTS

Biometric Face Detection & Recognition System — *Pure Java, AWT, Math Modeling* 2026

- Engineered an end-to-end real-time face detection and recognition pipeline from scratch in pure Java, strictly avoiding external Machine Learning or Deep Learning libraries to master core algorithmic mechanics.
- Implemented the Histogram of Oriented Gradients (HOG) algorithm from the ground up, extracting 8100-dimensional feature vectors using ITU-R BT.601 grayscale weights, gradient finite differences, and L_2 block normalization.
- Built a custom Support Vector Machine (SVM) classifier utilizing the Sequential Minimal Optimization (SMO) algorithm with a Sigmoid kernel, implementing a one-vs-all strategy for multi-person recognition.
- Designed a real-time sliding window detection system over camera frames utilizing Intersection over Union (IoU) based Non-Maximum Suppression (NMS) to accurately group and isolate overlapping bounding boxes.
- Developed a full-featured custom GUI using Java AWT for dataset capture, live camera streaming via OpenCV, preprocessing (rescaling to 128×128), and model training management entirely within a Linux environment.

PROFESSIONAL EXPERIENCE

Front-End Web Development Intern — *Pentalog* July 2024

- Built responsive, user-friendly web interfaces using HTML, CSS, JavaScript, and React, translating design mockups into functional, high-performance web components.
- Collaborated closely with cross-functional development teams in an Agile environment, gaining hands-on insight into modern software engineering workflows and best practices.

EDUCATION

"Alexandru Ioan Cuza" University of Iași — *Faculty of Mathematics* 2022 – 2026 (Thesis Pending)

- Bachelor of Science in Mathematics – Computer Science

"Ion Borcea" Theoretical High School, Buhuși 2018 – 2022

- Mathematics – Intensive Computer Science specialization.

TECHNICAL PROFICIENCIES

- **Programming Languages:** Python, Java, C++, C#, JavaScript, SQL, MATLAB, HTML/CSS
- **AI, LLM & ML Frameworks:** PyTorch, OpenCV, Scikit-Learn, NumPy, Pandas, LangChain, OpenAI API, Ollama
- **Web, Systems & Hardware:** React, Git, Linux Mint, Neovim / Vim, Arduino
- **Languages:** Romanian (Native), English (Professional Working Proficiency)